

Santu Sardar

RTL Design Engineer | FPGA | Verilog | RISC-V | AXI

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PROFESSIONAL SUMMARY

RTL Design Engineer with experience in **FPGA**-based system design and **SoC integration**, specializing in **AXI**-based interconnects, **DMA**-driven memory subsystems, and high-speed data paths. Proficient in **Verilog** RTL development, functional verification, and **FPGA validation** using Xilinx Vivado. Worked on **RISC-V** processors and **AI accelerator** architectures with a focus on memory and interconnect design. Holds an **M.Tech in VLSI Design** from NIT Surathkal.

EXPERIENCE

DIGITAL IC / SOC DESIGN ENGINEER

DeepGrid Semi Pvt. Ltd.

July 2025 – Present

Hyderabad, Telangana

- Implemented a ternary-weight **systolic matrix multiplier** for efficient AI inference and validated with it on FPGA (Arty A7-100T) using the **Microblaze** soft-core, executing **Bitnet MCU** MNIST inference via **Vitis C** firmware over **AXI** GPIO and UART.
- Integrated **DMA** and **AXI**-based data paths between QSPI Flash and HyperRAM.
- Collaborated with hardware teams to validate memory interfaces and ensure reliable high-speed data transfer, enabling autonomous model-weight loading independent of the RISC-V control CPU.

DIGITAL IC DESIGN INTERN

DeepGrid Semi Pvt. Ltd.

Jan 2025 – June 2025

Hyderabad, Telangana

- Developed Verilog RTL for **Softmax**, **ReLU** and **GeLU** modules for AI accelerator and performed functional verification using Xilinx Vivado.

PROJECTS

RISC-V 5-Stage Pipelined Processor | Verilog, Vivado

June 2024 – Dec 2024

- Designed and implemented a 5-stage pipelined RISC-V (RV32IM) processor with hazard detection, forwarding, and stalling mechanisms.
- Extended architecture to support floating-point operations (F-extension) for arithmetic instructions.
- Performed RTL design, simulation, and validation using Verilog and Xilinx Vivado.

Half-Precision Floating Point ALU | Verilog, Quartus Prime

Jan 2024 – May 2024

- Designed IEEE 754 compliant 16-bit floating-point ALU supporting addition, subtraction, multiplication, and division.
- Verified RTL modules in Verilog with functional validation in Intel Quartus Prime.

TECHNICAL SKILLS

Languages: Verilog HDL, SpinalHDL, C/C++.

Tools: Xilinx Vivado, Vitis, Intel Quartus Prime, Verilator, ModelSim.

Skills: RTL, Functional Verification, Testbench Development, Protocol Checking, Simulation Debugging, Waveform Analysis, Static Timing Analysis (STA), FIFO Design, Memory Systems, Digital VLSI Testing and Testability, Design Automation.

Protocols: AXI4, APB, SPI, Memory-Mapped Interfaces.

Domains: Digital Design, Design Verification, RTL Design, FPGA Design Engineer, DFT engineer, Memory Design engineer.

EDUCATION

National Institute of Technology Karnataka

Master of Technology in VLSI Design

Surathkal, Karnataka

Aug 2023 – June 2025

Jalpaiguri Government Engineering College

Bachelor of Technology in ECE

Jalpaiguri, West Bengal

Aug 2019 – May 2023

ACHIEVEMENTS

- Presented DgridSoC AI Accelerator work at the VLSI Design Conference 2026, Pune, showcasing system-level memory and interconnect architecture for AI inference workloads.
- Completed hands-on training on Electric Vehicle Systems using MATLAB/Simulink (NITTTR Chennai, Ministry of Education), covering system modeling, BMS diagnostics, CAN-based testing, and hardware deployment.
- Participated in a workshop on Hand Gesture Controlled Robot, gaining practical exposure to embedded systems and hardware interfacing.